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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : IT-605C

COMPILER DESIGN

Time Allotted : 3 Hours *Full Marks* : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP – A

(Multiple Choice Type Questions)

1. Choose the correct alternatives for the following : $10 \times 1 = 10$

i) Which of the task is performed by Lexical Analyser ?

a) Correcting errors

b) Stripping out comments and white spaces

c) Performs the expansion of macro

d) All of these.

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ii) Consider a grammar $A \rightarrow aS_1 \mid aS_2$. The left factored grammar produced from the grammar is

a) $A' \rightarrow aA$
$A \rightarrow S_1 \mid S_2$

b) $A \rightarrow aA'$
$A' \rightarrow aS_1 \mid aS_2$

c) $A \rightarrow a\,A'$
$A' \rightarrow S_1 \mid S_2$

d) none of these.

iii) Which of the following is a botton up parsing ?

a) LL parsing

b) Predictive parsing

c) Recursive decent parsing

d) Shift reduce parsing.

iv) Handle pruning forms the basis of

a) bottom up parsing

b) top down parsing

c) predictive parsing

d) recursive decent parsing.

v) Which of the following is not true for Syntax Directed Translation ?

a) It is an extension of CFG

b) Parsing process is used to do the translation

c) It does not permit the subroutines to be attached to the production of a CFG

d) It generates the intermediate code.

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vi) The reverse polish of suffix notation is known as

a) infix notation b) prefix notation

c) postfix notation d) none of these.

vii) Left factoring guarantees

a) not occurring of backtracking

b) cycle free parse tree

c) error free target code

d) correct LL(1) parsing table.

viii) Which of the following is not a loop optimization ?

a) Induction variable elimination

b) Loop jamming

c) Loop unrolling

d) Loop heading.

ix) Which data structure is used for shift-reduce parsing ?

a) Stack b) Queue

c) Array d) Pointer.

x) By which of the following bodies of two loops are merges to form a single loop ?

a) Loop unrolling

b) Strength reduction

c) Loop concatenation

d) Loop fusion.

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GROUP – B

(Short Answer Type Questions)

Answer any *three* of the following. $3 \times 5 = 15$

2. What is Syntax Directed Translation ? Define synthesized attribute, inherited attribute and explain with example. 2 + 3

3. Eliminate left recursion from the following grammar :

$E \rightarrow E + T \mid T$

$T \rightarrow TF \mid F$

$F \rightarrow F^* \mid a \mid b.$

4. What do you mean by ambiguous grammar ? Give example and explain why the grammar is ambiguous.

5. Give the following grammar

$S \rightarrow CC$

$C \rightarrow cC \mid d$

Construct the cannonical collection of LR (1) items and draw the state diagram for this grammar. 3 + 2

6. What is activation record ? Explain clearly its components. 2 + 3

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GROUP – C

(Long Answer Type Questions)

Answer any *three* of the following. 3 × 15 = 45

7. Consider the following grammar :

$E \rightarrow E + T \mid T$

$T \rightarrow T^* F \mid F$

$E \rightarrow (E) \mid id$

a) Construct SLR parsing table for this grammar.

b) Show the processing for the following expression using bottom-up parsing :

id + id * id . 10 + 5

8. a) Translate the following expression :

$a = b^* - c + b^* - c$

into

i) Quadruples

ii) Triples

iii) Indirect Triples.

b) What are the differences among quadruples, triples and indirect triples ?

c) Generate machine code for the following instruction :

$v = a + (b^* c) - d$. (3 + 3 + 3) + 3 + 3

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9. a) What is left factoring ?

b) Construct the LL(1) parsing table for the following grammar :

$S \rightarrow aBDh$

$B \rightarrow eC$

$C \rightarrow bc \mid \varepsilon$

$D \rightarrow g \mid \varepsilon$

c) Construct the SLR(1) parsing table for the following grammar :

$E \rightarrow E + T$

$E \rightarrow T$

$T \rightarrow T^* F$

$T \rightarrow F$

$F \rightarrow E$

$F \rightarrow id.$ 1 + 7 + 7

10. a) What is the difference between lexeme begin and forward pointer ?

b) Eliminate the left recursion from the following grammar :

$S \rightarrow (L) \mid a$

$L \rightarrow L, S \mid S$

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c) Find the FIRST an FOLLOW from the following production :

$S \rightarrow aBDh$

$B \rightarrow cC$

$C \rightarrow bC \mid \varepsilon$

$D \rightarrow EF$

$E \rightarrow g \mid \varepsilon$

$F \rightarrow f \mid \varepsilon$. 2 + 3 + 10

11. Write short notes on any *three* of the following : 3 × 5

a) Symbol table management

b) LEX tool

c) Cross compiler

d) Thomson's rule

e) Error recovery technique.

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